Exercise 1. Let X of finite dimensional vector space. Prove that any two norms on X are equivalent, that is, if $\|\cdot\|_1,\ \|\cdot\|_2$ are norms on X then there are positive constants A, B such that

$$A\|x\|_1 \leq \|x\|_2 \leq B\|x\|_1$$

for all $x \in X$. Conclude that the closed unit ball in a finite dimensional normed space is compact.

Exercise 2. Let X be a complex vector space and let $p : X \to \mathbb{R}_+$ be a function with the properties

$$p(x+y) \leq p(x) + p(y), \quad p(ax) = |a|p(x) \quad x, y \in X,\ \alpha \in \mathbb{C}.$$

Show that

$$Y = \{x \in X : \ p(x) = 0\}$$

is a subspace of X and that the map $q : X/Y \to \mathbb{R}_+$

$$q([x]) = p(x)$$

is well defined and is a norm on X/Y. Here $[x]$ denotes the coset of x in X/Y.

Exercise 3. Prove that if X is complete, $Y \subset X$ is a closed subspace, then X/Y is complete.

Exercise 4. Let X be a normed space and let Y be a proper linear subspace. Show that Y cannot have interior.

Exercise 5. Let X be a Banach space. Deduce from above that every proper closed subspace of X is of the first Baire category in X and use this to prove that the algebraic dimension of X is uncountable.

Exercise 6 Solve exercise 1 on page 53 in the book.

Exercise 7. Show that the norm on $l^p,\ 1 \leq p < \infty,\ p \neq 2$ does not come from a scalar product.